Process Characterization Plan

A-Mab Drug Substance — Ultrafiltration / Diafiltration (Step 10) — PCP-010

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

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Approvals

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Synthetic record. Signatures are not applied.

Revision history

Version	Effective date	Description of change
1.0	2026-02-23	Initial issue
2.0	2026-09-30	Reissued under INV-001: the rules of PCMP-001 Version 2.0 applied, execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
A280	protein concentration by ultraviolet absorbance
ALCOA+	attributable, legible, contemporaneous, original and accurate, and complete, consistent, enduring and available
AMD	analytical method decision
AMV	analytical method validation
BLA	Biologics License Application
CE-SDS	capillary electrophoresis with sodium dodecyl sulfate
CQA	critical quality attribute
DS	drug substance
DV	diavolume
HMW	high molecular weight species
HPLC	high performance liquid chromatography
IAC	intermediate acceptance criterion
icIEF	imaged capillary isoelectric focusing
MSAT	Manufacturing Science and Technology
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
PAR	proven acceptable range
psi	pounds per square inch
QA	Quality Assurance
SEC	size exclusion chromatography
SOP	standard operating procedure
TFF	tangential flow filtration
TMP	transmembrane pressure
UF/DF	ultrafiltration and diafiltration
UV	ultraviolet

1 Purpose and scope

This plan defines the process characterization studies for the ultrafiltration and diafiltration (UF/DF) step of the A-Mab drug substance process. UF/DF is Step 10, the last step of the drug substance train, and it delivers the drug substance at its target concentration of 75 g/L. The commercial process runs at a production bioreactor working volume of 15,000 L. The process is being transferred from the development site in Cambridge, MA to the commercial drug substance site in Grafton, WI. The studies of this plan form part of the Stage 1 process design record that supports the transfer and the Biologics License Application (BLA) (U.S. Food and Drug Administration 2011; International Council for Harmonisation 2008).

The plan covers the three process parameters of the step (Section 6.1), the scale-down model on which they are studied, the analytical methods and the sampling, the acceptance and decision criteria, and the analysis by which a proven acceptable range (PAR) is assigned to each parameter. It applies to the drug substance process only. Drug product manufacture and the development of the formulation belong to the drug product program and are out of scope. That program fixes the composition of the formulation buffer, and the studies treat it as a constant.

The step does not change the glycan variants of A-Mab, and no clearance of host cell protein, DNA or leached Protein A is credited to it (Section 4.2). It can add to aggregate and to the acidic charge variants (Section 4.1). No viral clearance is claimed for this step.

1.1 Changes in this revision

Version 1.0 of this plan was approved on 2026-02-23, before the first study of the step began on 2026-02-25. Version 2.0 is issued under the QA investigation INV-001, which concerned the acceptance criteria of the PAR analysis. QA opened INV-001 on 2026-08-03, after its review of the master report PCMR-001 Version 1.0 for BLA readiness, and closed it on 2026-09-18. The investigation found that the criteria used to judge each step in the PAR analysis had been set after the data were seen, and that no approved plan contained them (INV-001 finding 1). As corrective actions, the master plan PCMP-001 was reissued as Version 2.0 with acceptance criteria derived a priori (INV-001 corrective action CA1), and each step plan was amended with those criteria and with the method for fragments (corrective action CA2). This document is that amendment of PCP-010, issued after the studies it governs were run, and it describes only the studies that Version 1.0 planned. The design, the ranges, the levels and the sampling points of Version 1.0 are unchanged, and so are its decision rules except where the list below says otherwise. The changes are the following.

1. The criteria for aggregate, at most 5 % high molecular weight species (HMW), and for the acidic charge variants, 18 to 40 %, are the drug substance criteria, on the basis of PCMP-001 Version 2.0 (Section 7.2). PCMP-001 §7.3 defines 22 intermediate acceptance criteria (IACs) at six steps, for the step outputs on which a drug substance criterion depends. Each carries the drug substance criterion back through the platform worst case of every downstream step, which is the lowest clearance or the largest increase that the step showed for the earlier antibodies X-Mab, Y-Mab and Z-Mab. Version 1.0 of this plan stated the same criteria. At analysis, however, the Version 1.0 report PCR-010 read the acidic charge variants against their upper limit of 40 % alone (INV-001 finding 4). This plan keeps the two-sided criterion, and PCR-010 records the departure.

2. The rule of Version 1.0 for the scale-down model is replaced by the offset rule of PCMP-001 §7.1 (Section 5.1). Version 1.0 tested the step yield, the permeate flux and the change in aggregate for a difference from the pilot and commercial record at $\alpha = 0.05$, required the final concentration to lie within the variability of the A280 method, and assessed a significant difference for its size before rejecting the model (Section 7.1).

3. The statistics on which aggregate and the acidic charge variants are judged are named. Each is judged on single runs and not on a mean, as PCMP-001 §6.2 requires for a quality attribute (Section 7.2), and a slope is significant at $\alpha = 0.05$, the level of PCMP-001 §6.2 (Section 5.5). Version 1.0 judged single runs and tested slopes at the same level.

4. PCMP-001 Version 2.0 sets non-reduced CE-SDS (nrCE-SDS) as the method for fragments wherever a study measures them, following the analytical method decision AMD-001 of 2026-03-20 (PCMP-001 §7.8). The studies of this step do not measure fragments, a scope kept from Version 1.0 (Section 4.2).

5. The execution window of each study and the calibration status of the skid it used are given (Section 11). This answers INV-001 finding 5, that the Version 1.0 records gave no execution dates and the equipment tables therefore appeared to cite expired calibrations. The title block of Version 1.0 carried 2026-07-24, the date on which the Version 1.0 reports were approved. The revision history now gives, in its column for the effective date, the date on which Version 1.0 took effect, which is its approval date.

6. Two rules of PCMP-001 Version 2.0 are added to the classification (Section 7.4). Under PCMP-001 §6.2, “an attribute is called affected by a parameter when a term that contains the parameter is significant at that level”. Under PCMP-001 §7.7, “a significant effect that is too small to move the attribute appreciably within the PAR does not by itself make a parameter critical”. Version 1.0 did not define when a parameter affects an attribute, and it classified every parameter that affects one as critical or as well controlled critical.

7. Descriptions of the process that Version 1.0 gave wrongly are corrected, without a change to the design.

    - Version 1.0 said that the three parameters reach the product by different routes, and that beyond the flux limit a higher transmembrane pressure

(TMP) raises the antibody concentration at the membrane wall without raising throughput, as the final DS concentration does. The TMP and the final DS concentration share one route, the polarization layer. Below the flux limit a higher TMP drives more permeate and raises the wall concentration. Beyond it the flux no longer rises, the wall concentration is at its limit, and extra pressure compacts the layer at the wall. The final DS concentration acts on the flux, through the bulk concentration (Section 4.1, Section 4.3).

- Version 1.0 described the route to aggregate as mechanical and relied on shear. Shear alone causes little aggregation of an antibody. Interfaces, cavitation in the pump and particles shed by it are the usual causes, and the pump is not where the antibody is most concentrated (Section 4.1).
- Version 1.0 said that the mechanisms of the step do not depend on the sequence of the antibody. How readily an antibody aggregates depends on its colloidal stability and its self-association, which differ between antibodies (Section 4.1).
- Version 1.0 said that the viscosity of the retentate rises steeply as the antibody concentration approaches the DS target. This plan says only that it rises with concentration (Section 4.1).
- Version 1.0 said that the residual process buffer components fall exponentially with the number of diavolumes. They fall roughly exponentially, the conductivity approaches the value of the formulation buffer from above or below, and Donnan and volume-exclusion effects can leave the pH and the excipient levels of a concentrated retentate off the buffer values (Section 4.1).
- Version 1.0 said that the number of diavolumes sets how long the antibody is held. The final concentration and the TMP set it too (Section 4.1).
- Version 1.0 said that aggregate and the acidic charge variants are formed in the production bioreactor and that aggregate is reduced at the cation exchange step. Deamidation also continues wherever the antibody is held in solution, and aggregate also forms at the low-pH elution and hold (Section 4.2).
- Version 1.0 said that the membrane loading fixes the bulk concentration that the step reaches and the time the antibody spends on the skid. The final concentration is a parameter of the design, and the time differs between runs (Section 5.1).
- Version 1.0 said that the bench model passes the antibody through the pump more often per gram processed. With the crossflow and the membrane loading matched per unit area, the passes are the same at both scales. What differs is the hold-up, the share of the batch held in the recirculation loop, which is larger at bench scale, with a larger ratio of surface to volume (Section 5.1).
- Version 1.0 said that the conductivity and pH readings characterize the exchange curve and do not release a run. They decide the buffer-exchange criterion (Section 5.3).
- Version 1.0 said that neither an interaction nor a curvature term can be estimated from the design. Each series has three levels, so the statement is kept for interactions only (Section 5.5).

2 Objectives

The studies of this plan have the following objectives.

(i) To estimate the offsets of a bench-scale tangential flow filtration (TFF) model of the commercial skid from manufacturing scale, as the basis of its qualification (PCMP-001 §7.1).

(ii) To determine the effect of the number of diavolumes, the TMP and the final DS concentration on aggregate and on the acidic charge variants of the drug substance, over the ranges given in Section 6.1.

(iii) To establish the number of diavolumes at which the retentate reaches the formulation buffer, so that the lower edge of the operating range rests on data for A-Mab and not on platform practice alone.

(iv) To establish a PAR for each of the three parameters against the drug substance criteria for aggregate and the acidic charge variants (Table 4.1), by the analysis of Section 8.

(v) To provide the evidence on which the three parameters will be classified under SOP-4001, and to carry that classification into the control strategy through the report PCR-010 and the master report PCMR-001.

3 Related documents

This plan sits under the transfer plan PTP-001 and the master plan PCMP-001. It takes its study scope from the risk assessment RA-001, and its studies are reported in the report PCR-010 (Table 3.1).

Table 3.1: Related documents in the characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-010	Process Characterization Report (Ultrafiltration / Diafiltration (Step 10))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into
INV-001	Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review)	Reason for this revision

The procedures and validated methods that apply to the studies are listed in Table 3.2. The step is operated under SOP-2013, the scale-down model is qualified under SOP-1001, and the parameters are classified after the studies under SOP-4001.

Table 3.2: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
Analytical method validation	AMV-3011	Size-Variants (SEC-HPLC)
	AMV-3019	Protein Concentration by A280 (UV)
	AMV-3013	Charge Variants (icIEF)

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The UF/DF step concentrates the antibody and exchanges the process buffer for the formulation buffer. Both operations run across a tangential flow membrane whose pores pass buffer salts and retain the antibody. Ultrafiltration removes permeate and raises the antibody concentration in the retentate. Diafiltration then adds formulation buffer at the rate at which permeate leaves, so the retentate volume stays constant while the permeable species are washed out. Each diavolume removes a fixed fraction of the permeable species that remain, so the residual process buffer components fall roughly exponentially with the number of diavolumes. The conductivity of the retentate approaches the value of the formulation buffer from above or below. At high antibody concentration, Donnan and volume-exclusion effects can leave the pH and the excipient levels of the retentate off the values of the buffer. The step ends with a second concentration to the DS target of 75 g/L and a buffer flush that recovers the retentate.

The step can add to aggregate and to the acidic charge variants. The first route is physical. The membrane retains the antibody, so the antibody accumulates at the membrane wall in a polarization layer, where its concentration is higher than in the bulk retentate. The whole retentate also passes many times through the recirculation pump. Aggregate can form at the wall, where the antibody is most concentrated, and in the pump, where cavitation, air–liquid interfaces and particles shed by the pump can start it. Shear alone causes little aggregation of an antibody.

The A-Mab case study leaves the UF/DF step out (CMC Biotech Working Group 2009, 86), so it gives no expectation for this step. The expectations in the rest of this section come from general knowledge of the tangential flow filtration of antibodies. On that basis the risk of aggregate formation is expected to rise with the final concentration, with the TMP and with the number of passes through the pump. Below the point at which the polarization layer limits the flux, a higher TMP drives more permeate through the membrane and raises the antibody concentration at the wall. Beyond that point the flux no longer rises, the wall concentration is at its limit, and extra pressure compacts the layer at the wall. A higher crossflow raises the shear at the wall and lowers the antibody concentration there, because it sweeps retained antibody back into the bulk. A higher final concentration raises the viscosity of the retentate and, through the bulk concentration, lowers the flux. A higher viscosity raises the pressure drop along the feed channel. Within the normal operating ranges (NORs) the exposure is short and the effect is expected to be small.

The second route is chemical. The step keeps the antibody in solution for the length of the operation, and deamidation of asparagine residues continues during that time.

Deamidation raises the acidic charge variants. The increase grows with the time in solution, at a rate that rises with pH and temperature. The operation lasts longer with more diavolumes, with a higher final concentration, which lowers the flux, and with a lower TMP below the flux limit. For this reason the final retentate of every run is assayed for charge variants as well as for aggregate.

The step is part of the platform train that was used for X-Mab, Y-Mab and Z-Mab (PCMP-001 §2). The places where aggregate can form in the step are the same for every antibody, so that experience applies to the qualitative behaviour of the step. How readily an antibody aggregates depends on its colloidal stability and its self-association, which differ between antibodies, and the viscosity of its retentate, which rises with concentration, differs too. For this reason the final DS concentration is studied for A-Mab and is not carried over from the platform.

4.2 Quality attributes in scope

The studies measure two attributes of the drug substance across the step. Table 4.1 lists them with their drug substance criteria, the method that measures each in these studies, and the criticality assigned in the A-Mab attribute assessment (CMC Biotech Working Group 2009; International Council for Harmonisation 2023b).

Table 4.1: Quality attributes measured across the step, with their drug substance criteria.

Attribute	Criterion	Method	Criticality	Tool #1 score
Aggregates (HMW)	≤ 5 % HMW	SEC, AMV-3011	H	60
Acidic charge variants (deamidation)	18–40 %	icIEF, AMV-3013	VL	4

H, high criticality. VL, very low criticality. Tool #1 score, the impact of the attribute on safety and efficacy multiplied by the uncertainty of that impact (A-Mab attribute assessment).

Aggregate is the attribute most at risk at this step, from the high antibody concentration at the membrane wall and from interfaces and cavitation in the pump. The acidic charge variants are measured because deamidation continues while the antibody is in solution on the skid. Both attributes arrive with the load. They arise first in the production bioreactor. Deamidation continues wherever the antibody is held in solution, aggregate also forms at the low-pH elution and hold, and the cation exchange step reduces aggregate (PCMP-001 §5 and Table 7.1). The studies measure what this step adds to each, against the load from the virus filtration step.

The other attributes of the drug substance critical quality attribute (CQA) register are not measured across the step. Afucosylation, galactosylation and high mannose arise in the production bioreactor, and no downstream step changes them (PCMP-001 §3). The membrane retains every glycoform of the antibody, so the step cannot change their distribution. Host cell protein, residual DNA and leached Protein A are cleared by the chromatography steps before this one. Table 7.1 of the master plan

PCMP-001 credits this step with no clearance of host cell protein or leached Protein A, and residual DNA is judged on the drug substance only (PCMP-001 §7.3).

Fragments are not measured either. They form by proteolytic clipping of the antibody, in culture and during the harvest hold (PCMP-001 §3), and by non-enzymatic hydrolysis of the hinge region, which proceeds slowly with time in solution. The step adds no protease, and its physical route aggregates the antibody without cleaving it, but hinge hydrolysis continues on the skid. Version 1.0 left fragments out of the studies of this step, and this amendment keeps that scope. PCMP-001 assumes that this step does not change fragments (PCMP-001 Table 7.1), and the drug substance is tested for fragments at release.

Viral clearance is credited to the low-pH viral inactivation, anion exchange and virus filtration steps, and the cumulative claim is consolidated in the master report PCMR-001 (International Council for Harmonisation 2023a). It is not measured at this step.

4.3 Risk-based prioritization of parameters

The pre-characterization risk assessment RA-001 ranked every process parameter of the drug substance process for its potential impact on the CQAs and on process performance, and for its potential to interact with other parameters (International Council for Harmonisation 2023b). Parameters ranked high, or with a likely interaction, were assigned to multivariate study. Parameters ranked lower, whose range could be supported one factor at a time, were assigned to univariate study (PCMP-001 §4). All three parameters of this step were assigned to univariate study.

Each parameter acts on the product by its own route, and two of the routes meet in the polarization layer. More diavolumes wash more of the permeable process buffer components out of the retentate, and they lengthen the operation and add passes through the pump. A higher TMP raises the flux and the antibody concentration at the wall up to the flux limit, and beyond it compacts the layer at the wall. A higher final DS concentration raises the viscosity of the retentate and, through the bulk concentration, lowers the flux.

The final DS concentration therefore sets the flux limit, and with it the TMP at which the polarization layer begins to limit the flux. The effect of the TMP depends on the final DS concentration, and an interaction between them is possible. A study that varies one factor at a time cannot resolve such an interaction. What the design does bound is the effect of each parameter at the edge of its range studied, with the other parameters at their set-points (Table A.1). For an attribute that approaches its criterion at an edge, the rule of Version 1.0 is kept. Where a measured attribute approaches its acceptance criterion at an edge, the study should be extended to the combination of edges before a range is claimed, and the extension should be agreed with Statistics and QA before it is executed. The threshold for “approaches” is left to the report PCR-010, because Version 1.0 set none.

5 Materials and methods

5.1 Scale-down model and its qualification

The studies are run on the bench-scale skid EQ-TFF-142, registered as the ultrafiltration / diafiltration TFF skid and held under calibration and change control. Section 11 gives its calibration status for each study. The model holds the crossflow and the membrane loading at their commercial values, both scaled by membrane area (PCMP-001 §5). A higher crossflow per unit membrane area raises the shear at the membrane wall and, by sweeping retained antibody back into the bulk, lowers the antibody concentration in the polarization layer. The membrane loading, in grams of antibody per square metre, sets the volume that each square metre of membrane processes. At the same TMP and crossflow the flux is the same, so a run of the model lasts as long as the commercial step at the same settings. The membrane chemistry, the nominal molecular weight cut-off and the feed channel geometry are those of the commercial skid. The TMP, the number of diavolumes and the final concentration do not depend on scale and are set at the values under study.

One quantity cannot be matched, the share of the batch held up in the system. With the crossflow and the membrane loading matched per unit area, the antibody passes through the pump as often at bench scale as at commercial scale. The bench system, however, holds a larger share of its batch in its recirculation loop and has a larger ratio of surface to volume. More antibody is therefore lost to hold-up, and more of it meets surfaces and air–liquid interfaces per gram processed. Version 1.0 expected the bench model to be conservative for aggregate formation, because it assumed more passes through the pump at bench scale, and it expected a result that meets the criterion at bench scale to hold at commercial scale. The report PCR-010 compares these expectations with the data.

The qualification runs are made under SOP-1001, in study S-UFDF-SDM, where three runs at the set-point condition are compared with the pilot and commercial record of the step on the step yield, the permeate flux at the set-point TMP, the final concentration achieved and the change in aggregate content across the step. Under the master plan PCMP-001 §7.1 the offset of each output is the mean of the scale-down runs at set-point minus the mean of the manufacturing-scale batches, reported with its two-sided 90 % confidence interval. Each end of that interval is a one-sided 95 % confidence bound. PCMP-001 §7.1 requires the scale-down runs at set-point to be made, and the offset of each output to be estimated, before the first characterization run of the step. The decision whether the model is qualified for an output is made at analysis, because it needs the response-surface model of the step (Section 7.1). Version 1.0 did not name the manufacturing-scale batches of the comparison. PCR-010 names them and states when the offsets were estimated.

Table 5.1 shows how each output of the qualification is used. Of the outputs that the qualification compares, only aggregate carries a criterion at this step, the drug substance criterion for aggregate in the final retentate. Every judgement of a scale-down aggregate result against that criterion first subtracts the aggregate offset, as PCMP-001 §7.1 requires for an output with a criterion, so the conclusion is stated at manufacturing scale. PCR-010 states how the offset of the change across the step applies to the level in the final retentate.

Table 5.1: Outputs of the scale-down qualification and the use of their offsets.

Output	Compared at both scales	Criterion at the step	Use of the offset
Step yield	Yes	None	Recorded only
Permeate flux at the set-point TMP of 12 psi	Yes	None	Recorded only
Final concentration achieved	Yes	None	Recorded only
Change in aggregate content across the step	Yes	≤ 5 % HMW in the final retentate	Subtracted before every judgement
Acidic charge variants	No	18–40 % in the final retentate	None, the offset is unknown
Buffer exchange (conductivity and pH)	No	Plateau of the formulation buffer at or below 7 DV	None, the offset is unknown

The acidic charge variants and the buffer exchange also carry criteria at this step. The qualification does not compare them at both scales, so their offsets are unknown, and both are judged as the scale-down model measures them (PCMP-001 §7.1). For the acidic charge variants this rests on the kinetics of deamidation, which depend on the pH and the temperature of the solution and on the time spent in it, and not on the scale of the skid. The model runs under the operation procedure of the commercial step (SOP-2013) in the commercial formulation buffer. For the buffer exchange it rests on the passage of the buffer species through the membrane, which depends on the number of diavolumes and on the membrane and not on scale.

5.2 Operation

The step receives the filtrate pool of the small virus retentive filtration step. The pool is concentrated by ultrafiltration, diafiltered at constant volume against formulation buffer for the planned number of diavolumes, concentrated a second time to the final target, and recovered with a buffer flush. Operation follows SOP-2013. At the set-point condition the step runs 8 DV at a TMP of 12 psi to a final concentration of 75 g/L.

All runs of the study are loaded from a single pool of the virus filtration step, so that the variability of the load does not confound the comparison between the levels

of a parameter. Section 12 gives the bridging rule that applies if one pool is not sufficient.

5.3 Analytical methods

The validated methods are listed in Table 3.2. Aggregate is measured by size exclusion chromatography (SEC) under AMV-3011 and reported as the percentage of HMW species. SEC remains the method for aggregate under the master plan PCMP-001 §7.8. Protein concentration is measured by ultraviolet absorbance (A280) under AMV-3019. The mass balance and the final concentration rest on that measurement. Charge variants are measured by imaged capillary isoelectric focusing (icIEF) under AMV-3013 and reported as the percentage of acidic species.

The completeness of the buffer exchange is followed by the conductivity and the pH of the retentate. These are in-process readings from the skid instrumentation, not validated assays. They decide the buffer-exchange criterion of Section 7.2, and so whether the lower edge of the diavolume NOR rests on data for A-Mab.

Aggregate binds the cation exchange resin more strongly than the monomer, so it elutes late and is largely excluded from the pool at the cation exchange step. The aggregate content of A-Mab drug substance is low, and a change across this step may be of the same size as the repeatability of the SEC method. The rules of Version 1.0 for such small changes are kept. Analytical Development will report the method variability that applies over the range in which the samples fall. A measured change is judged against the pure error estimate described in Section 5.5 rather than against the nominal precision of the method. Noise near the low end of the method range will be attributed to the assay rather than to the parameter under study, unless the change measured across the range is larger than the pure error estimate of Section 5.5. How this last rule and the slope test of Section 5.5 are applied together is stated in the report PCR-010, because Version 1.0 did not relate them.

5.4 Sampling plan

Samples will be taken at the points listed in Table 5.2. The acceptance criteria for aggregate and the acidic charge variants (Section 7.2) apply to the final retentate, and the buffer-exchange criterion to the diafiltration series. The other samples support the mass balance and the confirmation that the membrane retains the antibody.

Table 5.2: Sampling plan of the characterization runs.

Sample	Point in the step	Tests	Purpose
Load pool	Virus filtration filtrate, before ultrafiltration	A280, SEC, icIEF	Measures the input to every run in the study
Pre-diafiltration retentate	End of the first concentration	A280	Confirms the concentration at which diafiltration starts
Diafiltration series	After each diavolume, in the runs of the diavolume series	Conductivity, pH [a)]	Builds the buffer exchange curve, on which the buffer-exchange criterion is read
Final retentate	After the final concentration and recovery flush	A280, SEC, icIEF	The drug substance sample to which the criteria for aggregate and acidic variants apply
Permeate	Composite over the run	A280	Confirms that the membrane retains the antibody

[a)] Read from the skid instrumentation. In-process readings, not validated assays.

Every sample will be assayed within the stability period established for the matrix, and the load pool sample will be assayed once for the whole study rather than once per run. Where a run is repeated, the repeat carries its own sample set.

5.5 Statistical methods

For each parameter and each response, the measured values will be regressed on the level of the parameter over the range studied, and the slope will be tested against zero at $\alpha = 0.05$. The set-point condition is shared by the three series and will be run three times. Those replicates give the pure error estimate of the combined process and assay variability, against which a change across a range is read. The master plan PCMP-001 §6.2 sets the same level for the significance of a model term, and under it an attribute is called affected by a parameter when the slope is significant at that level.

Statistical significance and practical significance will be reported separately. A slope that is significant at $\alpha = 0.05$ but predicts a change across the range that is small against the acceptance criterion will be described in those terms, and the criterion rather than the p-value decides whether the range is acceptable. No response-surface model will be fitted for this step, because the design carries no factorial structure and no interaction term can be estimated from it. The model-adequacy criteria of PCMP-001 §7.2 apply to response-surface models and therefore do not apply here.

The buffer exchange will be analysed separately. The conductivity of the retentate will be plotted against the number of diavolumes, and the exponential decay expected

from Section 4.1 will be fitted to the series. The analysis reports the number of diavolumes at which the conductivity of the retentate has reached the plateau set by the formulation buffer and no further change is measured over the following diavolume.

6 Study design

6.1 Factors, ranges and study type

The parameters, their set-points, the ranges to be studied and the NORs are given in Table 6.1. All three are studied one at a time, because the risk assessment RA-001 assigned them to univariate study (Section 4.3).

Table 6.1: Parameters, ranges studied and normal operating ranges for the step.

Parameter	Unit	Set-point	Range studied	NOR
Number of diavolumes	DV	8	7–10	7–9
Transmembrane pressure	psi	12	10–15	10–14
Final DS concentration	g/L	75	65–85	70–80

Each range studied extends beyond the NOR at one end or at both. The number of diavolumes is studied from 7 to 10 DV against a NOR of 7 to 9 DV. The range extends above the NOR because more diavolumes mean more time on the skid and more passes through the pump. Its lower edge is the lower edge of the NOR, which is the lowest count the commercial step will run, and the buffer exchange curve characterizes the step at that value. Below that edge the buffer exchange would be less complete, and the study does not go there. The TMP is studied from 10 to 15 psi against a NOR of 10 to 14 psi. The range extends above the NOR because pressure above the flux limit compacts the polarization layer without raising throughput, so more of the antibody sits at the highest concentration at the wall. Its lower edge is also the lower edge of the NOR. A lower TMP would lengthen the operation, and with it the passes through the pump and the time for deamidation, and the study does not characterize that side beyond the NOR.

The final DS concentration is studied from 65 to 85 g/L against a NOR of 70 to 80 g/L, and the range extends on both sides. An excursion above the target raises the viscosity and the bulk concentration of the retentate, lowers the flux and lengthens the operation, and so raises the risk of aggregation. An excursion below the target leaves the drug substance below the concentration that the drug product program expects.

The step will not be characterized outside these ranges, and no range wider than the range studied will be proposed in the report PCR-010.

6.2 Univariate assessment

Each parameter is run at three levels, the two edges of its range studied and its set-point, with the other two parameters held at their set-points. The set-point condition is common to the three series, so the study consists of six edge runs and three replicates of the set-point condition, nine runs in all. The planned runs are listed in Appendix A (Table A.1).

The order of execution will be randomized across the whole study. Each run is sampled at the points of Table 5.2, and its final retentate is assayed by SEC, A280 and icIEF. The diavolume series carries the additional in-process sampling that builds the buffer exchange curve.

7 Acceptance and decision criteria

7.1 Scale-down model

Version 1.0 accepted the model under a rule in three parts. The mean step yield, the permeate flux and the change in aggregate content across the step were not to differ from the pilot and commercial record at $\alpha = 0.05$. The final concentration achieved was to lie within the reported variability of the A280 method. A difference that was statistically significant was to be assessed for its size against the acceptance criteria of Section 7.2 before the model was rejected, and that judgement was to be documented in the qualification record and approved by Statistics and QA.

The master plan PCMP-001 Version 2.0 replaces such tests with the offset of each output from manufacturing scale (PCMP-001 §7.1, Section 5.1). Under PCMP-001 a model is qualified for an output when the conclusion over the NOR is the same with the offset at either end of its confidence interval, and that conclusion is drawn from the response-surface model of the step. This step has no response-surface model, so the qualification decision cannot be made as PCMP-001 states it, and this plan sets no rule for it. The report PCR-010 states the approach it takes.

7.2 Product quality

The final retentate is judged against the drug substance criteria of Table 4.1. Aggregate must not exceed 5 % HMW, and the acidic charge variants must lie between 18 and 40 %. The master plan PCMP-001 §7.3 carries each drug substance criterion back through the platform worst cases of the downstream steps to give the IACs of the steps before this one. No step follows this one, so PCMP-001 Table 7.1 defines no IAC for it, and the drug substance criteria apply directly (PCMP-001 §7.6). They come from the drug substance CQA register (PCMP-001 §3), and no result of the studies of this plan enters them.

Under PCMP-001 §6.2 a quality attribute is judged on a single batch, not on a mean. At this step each run is judged on its own measured result.

The derivation in PCMP-001 assumes that no step after cation exchange changes aggregate, and that no step after low-pH viral inactivation changes the acidic charge variants (PCMP-001 Table 7.1). This step can add to both, and the studies of this plan measure what it adds. The verification of platform worst cases in PCMP-001 §7.5 uses the response-surface model of a step, and no verification is planned at this step.

The buffer-exchange criterion is that of Version 1.0. The buffer exchange is accepted when the conductivity and the pH of the retentate have reached the plateau of the formulation buffer, by the analysis of Section 5.5, at or below the lower edge of the diavolume NOR of 7 DV. The analysis of Section 5.5 treats the conductivity only. Version 1.0 set no analysis for the pH and no numerical tolerance for the plateau, so the report PCR-010 states its approach to both.

7.3 Decision rules

Version 1.0 set the following decision rules, and they are unchanged.

1. A level of a parameter is acceptable when every attribute measured on the final retentate of that run meets its criterion.
2. A parameter passes over its range studied when every level in the range is acceptable and the change measured across the range is small against the criterion.
3. Where a level fails, the range is truncated at the last acceptable level, and the failing level is reported as an edge of failure.
4. Where a failure occurs at a level inside the NOR, execution stops, and the operating range is reviewed with the receiving site and with QA before the study continues.
5. Where a run fails for a reason attributable to the equipment or to the method, and not to the parameter under study, the run may be repeated once. Both the original and the repeat are reported in PCR-010.

The threshold for a change that is small against the criterion is left to PCR-010, because neither Version 1.0 nor PCMP-001 sets one.

7.4 Classification

Each parameter will be classified under SOP-4001 (International Council for Harmonisation 2009, 2012). The classification rule of Version 1.0 is kept. A parameter whose variation over the characterized range affects no quality attribute, but whose control is needed for the step to perform consistently, is classified as a key process parameter. A parameter whose variation affects a quality attribute is classified as critical, or as well controlled critical where the risk of falling outside the design space is low. The classification is an outcome of the studies and is reported in PCR-010.

At a step with no response-surface model, the term that the master plan PCMP-001 §6.2 tests is the slope of Section 5.5, and the effect that PCMP-001 §7.7 weighs is the change that this slope predicts across the PAR. A well controlled critical parameter is one that is readily controlled and whose PAR is wide against its NOR (PCMP-001 §7.7), and a parameter studied one factor at a time is classified by the same rules as one studied in a design (PCMP-001 §4). PCMP-001 and SOP-4001 set no numerical threshold for a small effect or a wide PAR, so the report PCR-010 states the size of

each effect across the PAR and the margin of the PAR to the NOR, and classifies on them.

8 Proven acceptable ranges (planned analysis)

A PAR will be assigned to each parameter from its univariate series. The acceptance basis is the drug substance criteria for aggregate and the acidic charge variants (Table 4.1). The PAR of a parameter is the contiguous span of levels over which every attribute measured on the final retentate met its criterion, with the other parameters at their set-points. Each PAR will be reported in the report PCR-010 together with the measured series and the criterion against which it was read.

Three limitations apply to a PAR determined this way, and PCR-010 will state them where it states the ranges.

1. A univariate PAR holds with the other parameters at their set-points. It does not describe the step when two parameters sit at their range edges at the same time.
2. The master plan PCMP-001 §6.4 defines a second analysis, the NOR-propagated analysis, which predicts the response from a fitted multivariate model while the other parameters are drawn at random around their set-points. This step has no such model, so that analysis is not performed. PCMP-001 §7.6 defines a PAR through it, and for a parameter studied one factor at a time the range rests on its own data instead (PCMP-001 §4).
3. A PAR cannot be wider than the range studied. Where every level in a range is acceptable, the claimed PAR is the range studied of Table 6.1 and nothing beyond it, and it is reported as reaching the edge of that range (PCMP-001 §7.6).

The ranges reported under this analysis become the basis of the operating ranges that the receiving site runs to, together with the classification of Section 7.4. Where a PAR is wider than the NOR, the margin between them is what a deviation is dispositioned against. Where a PAR is truncated at an edge of failure, the NOR should be reviewed against the truncated range before the step is transferred.

9 Data management and integrity

Data generated under this plan are recorded so that they are attributable, legible, contemporaneous, original and accurate, and so that they remain complete, consistent, enduring and available (ALCOA+). Run records are captured in the electronic notebook under the study number. Analytical results are captured in the laboratory information management system and reported against the sample identifiers of each run. As INV-001 preventive action PA3 requires, the study record also gives the execution dates of each study and the calibration status of the equipment it used.

The statistical analysis of Section 5.5 is scripted and held under version control, so that it can be re-executed on the retained raw data and return the same result. Every calculation is reviewed by a second qualified person. A change to a record after it has been made carries an audit trail with the reason for the change. Values quoted in the report PCR-010 are read from the retained data set and are not re-entered by hand. Records are retained under the record retention procedure of the sponsor for the life of the product.

10 Roles and responsibilities

The functions responsible for the study and their responsibilities are listed in Table 10.1. Process Development owns the study. Quality Assurance approves this plan before any run is executed. This Version 2.0 is the amendment that INV-001 required of each step plan as a corrective action, issued after the runs and approved on its effective date, 2026-09-30.

Table 10.1: Functions responsible for the study.

Function	Responsibility
Process Development	Designs the study, executes the runs, analyses the data and drafts PCR-010
Manufacturing Science and Technology	Owns the scale-down model and its qualification, and carries the outcome into the transfer
Analytical Development	Releases the methods, runs the assays and reports the method variability
Statistics	Reviews the design before execution and the analysis before the report is approved
Quality Assurance	Approves this plan and PCR-010, and dispositions any deviation from the plan
Process science at the receiving site	Reviews the resulting ranges against the commercial skid and confirms that they are achievable

The receiving site takes part in the review of this plan and of the report PCR-010. The ranges that the studies support are thus assessed against the commercial skid before they are written into the control strategy (International Council for Harmonisation 2008).

11 Deliverables and schedule

The deliverable of this plan is the report PCR-010, the process characterization report for the step. PCR-010 reports the qualification of the scale-down model, the three univariate series, the buffer exchange curve, the PARs and the parameter classification, and it rolls up into the master report PCMR-001.

The work runs in the following order.

1. The qualification runs of the scale-down model.
2. The characterization runs of Section 6.2, from a single load pool.
3. The analysis of Section 5.5 and the PAR determination of Section 8, after the last run.
4. The drafting, review and approval of PCR-010.
5. The transfer of the classification and the ranges into the control strategy and into PCMR-001.

Version 1.0 gave no calendar dates. The studies of this plan ran in the windows of Table 11.1, on the skid EQ-TFF-142, whose next calibration was due on 2026-04-22. A window runs from the first to the last day of execution in the study record. The qualification study ended before the univariate study began, and both windows closed before the calibration due date. The campaign schedule is held in the master plan PCMP-001 and the transfer milestones in the transfer plan PTP-001.

Table 11.1: Execution windows of the studies of this plan.

Study	Study type	Start	End
S-UFDF-SDM	Scale-down model qualification	2026-02-25	2026-03-06
S-UFDF-UNI	Univariate	2026-03-09	2026-03-27

12 Risks and assumptions

A failure of aggregate at bench scale is handled by a rule of Version 1.0, which is kept. If aggregate fails its criterion at bench scale, the condition will be repeated at pilot scale before an operating range is narrowed on the strength of the bench result.

If one pool is not sufficient for the whole study, a second pool will be bridged by running the set-point condition on both pools and comparing the responses against the pure error estimate of Section 5.5. A difference larger than the pure error will be investigated before the second pool is used, and the runs executed on each pool will be identified in the report PCR-010.

The design has one run at each edge of a range. The slope test of each series therefore rests on few runs, and an effect that is small against the variability between runs may not reach significance.

The plan rests on the following assumptions.

1. The composition of the formulation buffer is fixed by the drug product program. A change to the formulation would change the target of the buffer exchange and would invalidate the diavolume study.
2. The membrane chemistry and the membrane supplier are as qualified, and a change to either requires a new risk assessment before these data are applied.
3. The material entering the step arrives within the criteria that the master plan PCMP-001 sets for the output of the virus filtration step (PCMP-001 Table 7.1). The study characterizes the step under representative load.

13 Approvals

This plan takes effect when it has been approved by the functions recorded in the Approvals table at the front of this document. A change to the design, to the ranges or to the acceptance criteria after approval requires a revision of this plan, approved by the same functions, before the affected runs are executed. A departure from the plan during execution will be recorded as a deviation, dispositioned by QA, and reported with its impact assessment in the report PCR-010. Work that the plan did not foresee, and that departs from no approved procedure, is recorded in PCR-010 as a discrepancy from this plan.

Appendix A. Planned univariate runs

Table A.1 lists the settings of the planned runs, series by series. No coded design matrix is given for this step, because the study has no factorial design.

Table A.1: Settings of the planned runs of the univariate study.

Series	Level	Diavolumes (DV)	TMP (psi)	Final DS concentration (g/L)
Number of diavolumes	Low edge	7	12	75
	Set-point	8	12	75
	High edge	10	12	75
Transmembrane pressure	Low edge	8	10	75
	Set-point	8	12	75
	High edge	8	15	75
Final DS concentration	Low edge	8	12	65
	Set-point	8	12	75
	High edge	8	12	85

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